

Q1 - 25 July - Shift 1

A solid cylinder and a solid sphere, having same mass M and radius R , roll down the same inclined plane from top without slipping. They start from rest. The ratio of velocity of the solid cylinder to that of the solid sphere, with which they reach the ground, will be :

(A) $\sqrt{\frac{5}{3}}$

(B) $\sqrt{\frac{4}{5}}$

(C) $\sqrt{\frac{3}{5}}$

(D) $\sqrt{\frac{14}{15}}$

Space for your notes:

Q2 - 26 July - Shift 1

A disc of mass 1 kg and radius R is free to rotate about a horizontal axis passing through its centre and perpendicular to the plane of disc. A body of same mass as that of disc is fixed at the highest point of the disc. Now the system is released, when the body comes to the lowest position, its angular speed will be

$$4\sqrt{\frac{x}{3R}} \text{ rad s}^{-1} \text{ where } x = \underline{\hspace{2cm}}$$

$$(g = 10 \text{ ms}^{-2})$$

Space for your notes:

Q3 - 26 July - Shift 2

Questions

MathonGo

The radius of gyration of a cylindrical rod about an axis of rotation perpendicular to its length and passing through the center will be _____ m.

Given, the length of the rod is $10\sqrt{3}$ m.

Space for your notes:

Q4 - 27 July - Shift 1

A pulley of radius 1.5 m is rotated about its axis by a force $F = (12t - 3t^2)$ N applied tangentially (while t is measured in seconds). If moment of inertia of the pulley about its axis of rotation is 4.5 kg m^2 , the number of rotations made by the pulley before its direction of motion is reversed, will be $\frac{K}{\pi}$. The

value of K is _____.

Space for your notes:

Q5 - 28 July - Shift 1

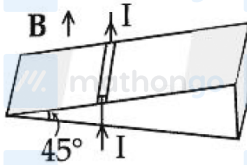
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Questions

MathonGo

As shown in the figure, a metallic rod of linear density 0.45 kg m^{-1} is lying horizontally on a smooth incline plane which makes an angle of 45° with the horizontal. The minimum current flowing in the rod required to keep it stationary, when 0.15 T magnetic field is acting on it in the vertical upward direction, will be :

{Use $g = 10 \text{ m/s}^2$ }



- (A) 30 A (B) 15 A
(C) 10 A (D) 3 A

Space for your notes:

Q6 - 29 July - Shift 2

The torque of a force $5\hat{i} + 3\hat{j} - 7\hat{k}$ about the origin is τ . If the force acts on a particle whose position vector is $2\hat{i} + 2\hat{j} + \hat{k}$, then the value of τ will be :

- (A) $11\hat{i} + 19\hat{j} - 4\hat{k}$ (B) $-11\hat{i} + 9\hat{j} - 16\hat{k}$
(C) $-17\hat{i} + 19\hat{j} - 4\hat{k}$ (D) $17\hat{i} + 9\hat{j} + 16\hat{k}$

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Answer Key

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Q1 (D)

Q2 (5)

Q3 (5)

Q4 (18)

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Q5 (A)

Q6 (C)

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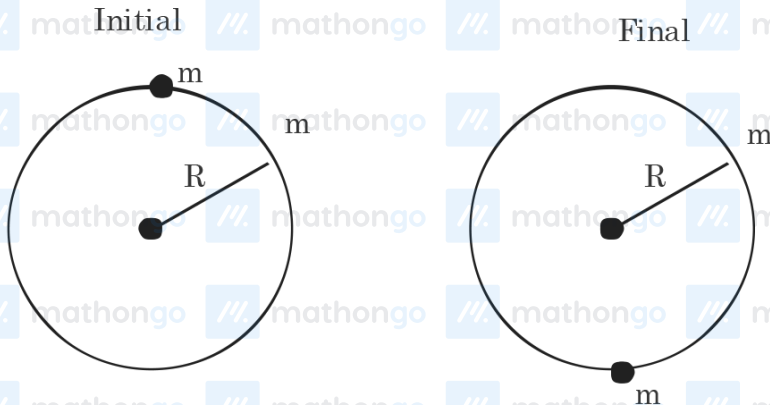
Q1 (D)

$$V = \sqrt{\frac{2gH}{1 + k^2 / R^2}}$$

$$\frac{V_{\text{cylinder}}}{V_{\text{sphere}}} = \sqrt{\frac{(1 + k^2 / R^2)_{\text{sphere}}}{(1 + k^2 / R^2)_{\text{cylinder}}}}$$

$$= \sqrt{\frac{1 + 2/5}{1 + 1/2}} = \sqrt{\frac{7}{5} \times \frac{2}{3}} = \sqrt{\frac{14}{15}}$$

Q2 (5)



using conservation of mechanical energy

$$mg2R = \frac{1}{2} I_{\text{disc}} \omega^2 + \frac{1}{2} I_{\text{particle}} \omega^2$$

$$mg2R = \frac{\omega^2}{2} \left[\frac{mR^2}{2} + mR^2 \right]$$

$$mg2R = \frac{\omega^2}{2} \frac{3}{2} mR^2$$

$$\frac{3}{4} \omega^2 = \frac{2g}{R}$$

$$\omega^2 = \frac{8g}{3R}$$

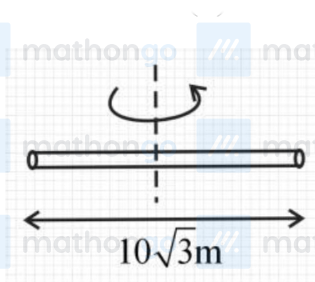
$$\omega = \sqrt{\frac{80}{3R}}$$

Given $\omega = 4\sqrt{\frac{x}{3R}}$

$$16 \frac{x}{3R} = \frac{80}{3R}$$

$$x = 5$$

Q3 (5)



$$I = \frac{m\ell^2}{12} = mk^2 \Rightarrow k^2 = \frac{\ell^2}{12} \Rightarrow k = \frac{\ell}{\sqrt{12}} = \frac{\ell}{2\sqrt{3}} = \frac{10\sqrt{3}}{2\sqrt{3}} = 5$$

Q4 (18)

$$\tau = I \alpha \Rightarrow (12t - 3t^2)1.5 = 4.5 \alpha$$

$$\Rightarrow \alpha = 4t - t^2$$

$$\Rightarrow \frac{d\omega}{dt} = 4t - t^2 \Rightarrow \omega = \int_0^t (4t - t^2) dt$$

$$\Rightarrow \omega = 2t^2 - \frac{t^3}{3}$$

$$\text{For } \omega = 0 = 2t^2 - \frac{t^3}{3} \Rightarrow t^2 \left(2 - \frac{t}{3} \right) = 0$$

$$\Rightarrow t = 0, 6.$$

$$\frac{d\theta}{dt} = 2t^2 - \frac{t^3}{3} \Rightarrow \theta = \int_0^6 \left(2t^2 - \frac{t^3}{3} \right) dt$$

$$= \left[\frac{2t^3}{3} - \frac{t^4}{12} \right]_0^6$$

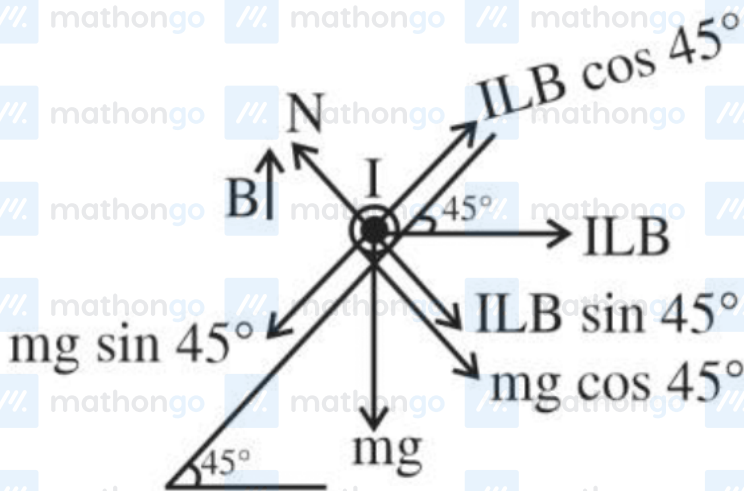
$$= 6^3 \left(\frac{2}{3} - \frac{6}{12} \right) = 6^3 \left(\frac{8-6}{12} \right)$$

$$= \frac{6^3}{6} = 36$$

$$\text{No. of revolutions} = \frac{36}{2\pi} = \frac{18}{\pi}$$

$$\therefore K = 18$$

Q5 (A)



$$mg \sin 45^\circ = ILB \cos 45^\circ$$

$$\therefore I = \left(\frac{m}{L} \right) \frac{g}{B}$$

$$= \frac{(0.45)(10)}{0.15} = 30 \text{ A}$$

Q6 (C)

$$\vec{\tau} = \vec{r} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 2 & 1 \\ 5 & 3 & -7 \end{vmatrix}$$
$$= \hat{i}(-14 - 3) - \hat{j}(-14 - 5) + \hat{k}(6 - 10)$$
$$= -17\hat{i} + 19\hat{j} - 4\hat{k}$$